

Solar Resource & Economics Report

Phoenix AZ · Austin TX · Charlotte NC · Trailing 365-Day Real Irradiance Data

✓ **REAL DATA** — irradiance figures below are pulled live from Open-Meteo's public archive API at generation time (trailing 365 days: 2025-08-10 to 2026-08-10), not synthetic or hardcoded. Production/economics figures apply disclosed formulas and assumptions to that real data — see the Methodology section on page 2.

SOLAR RESOURCE — TRAILING 365-DAY AVERAGE

19.9 BEST MARKET: PHOENIX, AZ MJ/M²/DAY	5.5 PEAK SUN HOURS/DAY (BEST MARKET)	10.8 FASTEST SIMPLE PAYBACK (YRS) — PHOENIX, AZ
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MARKET	AVG MJ/M²/DAY	PEAK SUN HOURS/DAY	EST. ANNUAL KWH (7.6KW SYSTEM)
Phoenix, AZ	19.9	5.5	12,261
Austin, TX	18.0	5.0	11,112
Charlotte, NC	17.1	4.8	10,561

Phoenix, AZ leads all three markets at 19.9 MJ/m²/day average irradiance, translating to an estimated 12,261 kWh/year for a 7.6kW DC system. This is real trailing-year data for each location, not a seasonal snapshot or a single sunny day.

Irradiance data: Open-Meteo archive API (open-meteo.com), free/no-key, ERA5-based reanalysis, trailing 365 days as of report generation (2025-08-10 to 2026-08-10). Production estimates apply a disclosed methodology — see page 2. Not a substitute for a site-specific solar assessment, which accounts for roof orientation, shading, and local permitting. · Prepared by Mboya Jeffers · mboyajeffers9@gmail.com · Generated August 12, 2026

ESTIMATED ECONOMICS BY MARKET

MARKET	EST. ANNUAL VALUE	EST. SYSTEM COST	SIMPLE PAYBACK
Phoenix, AZ	\$1,962	\$21,280	10.8 yrs
Austin, TX	\$1,778	\$21,280	12.0 yrs
Charlotte, NC	\$1,690	\$21,280	12.6 yrs

⚠ **ASSUMPTIONS** (disclosed, not real-time) — 7.6kW DC system size · 80% system-loss derate factor · \$0.16/kWh electricity rate (approx. US residential average — confirm your local utility rate) · \$2.80/watt gross install cost (pre-incentive). Federal/state/utility incentives vary by jurisdiction and change over time — confirm current eligibility with a tax professional before relying on any payback figure. Not tax or financial advice.

METHODOLOGY

Real data: daily shortwave radiation sum (MJ/m²/day) for each market's coordinates, trailing 365 days, from Open-Meteo's free archive API — no API key required, no synthetic fallback (the script fails loudly if the source is unreachable rather than substituting placeholder data).

Peak sun hours: MJ/m²/day ÷ 3.6 = kWh/m²/day, numerically equivalent to peak sun hours (1 kW/m² is the standard test-condition irradiance used industry-wide).

Annual production estimate: System size (kW DC) × peak sun hours/day × 365 × derate factor. The derate factor is a standard, disclosed assumption representing inverter efficiency, wiring loss, soiling, temperature de-rating, and shading — a conservative estimate in line with common industry rules of thumb (NREL's PVWatts tool uses a similar but not identical default; a site-specific assessment would use your actual roof/shading conditions).

Payback period: Estimated system cost (gross, pre-incentive) ÷ estimated annual production value. This is a simple payback calculation, not a full lifecycle NPV/IRR model, and does not include financing costs, degradation over time, or electricity rate escalation.

WHAT THIS SYSTEM GENERATES FROM YOUR DATA

This report runs the same real-data methodology against any location and any system size. Given your actual roof size/orientation, local electricity rate, and current local incentive eligibility, it produces a site-specific version of this same analysis.

To generate this report for your specific site: share your address (or coordinates) and rough roof size — the first pass is free, no commitment. **Turnaround: 48 hours.**

Irradiance data real and live at generation time (Open-Meteo archive API). Production and economics figures apply the disclosed assumptions above to that real data. Not a substitute for a licensed installer's site assessment. Not tax or financial advice — confirm current federal/state/utility incentive eligibility with a qualified professional. · Prepared by Mboya Jeffers · mboyajeffers9@gmail.com · Generated August 12, 2026